Which Way Does the Condition Point?

Unit 2 · Topic 2.6 · TODAY'S PROBLEM

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

Mina: "The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it."

Luis: "We need to count everyone who tested positive before finding the chance of having the condition."

Screening results (counts)

Status	Pos.	Neg.	Total
Condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

FIRST TAKE (5 min, on your sheet)

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

DOK 1 **(a)** For $P(\text{condition} \mid \text{positive})$, identify the numerator and denominator counts.

DOK 2 **(b)** Find $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Show each count fraction and percent.

DOK 3 **(c)** Is Mina's claim supported? Use the positive-test counts for students with and without the condition. Explain how reversing given changes the group in the denominator. Write 3–4 sentences.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.